Jake Quackenbush

CSC220 – Intro to Machine Learning

Lab 2 – Regression Models

Dataset Selection: “`sklearn.datasets import load_diabetes`”

Models: Gradient Boosting, Linear Regression, Random Forest, Polynomial, Elastic Net

Model Selection Justification

- Linear Regression and Gradient Boosting both shine with more linear datasets which could have provided a large boost with this dataset. The Random Forest appeared to have the most potential at first, the R^2 shows us that the model was overfit for the dataset and began to memorize instead of processing the data effectively
- The results of the post-hyperparameters were interesting. The values for MAE and MSE grew dramatically but the gaps between the training and testing data shrunk. It still showed signs of overfitting but was the most interesting to tune and tweak by far. The fitting of linear regression and gradients suggests that the data has a linear lean to it and might do better with those types of models.

Feature Analysis

- BMI was consistently the most important feature across my models. The runners-up were Blood Pressure (bp) and LDL (s2).
- BMI stayed consistent across almost all the models, anything past s2, bp, and s5 varied by such small differences that they averages are all so close together that they seem very insignificant compared to the main 4 features.
- Find and exploit the more linear combinations within the data to really maximize the effectiveness of something like a Linear Regression or Gradient Boosting model.

Practical Considerations

- Following the EDA and initial testing I think that the Gradient or Random Forest may have been the best in production. The Gradient Boost had some of the most accurate default scores but did not receive further testing. The random forest was for more generalized but overfit to the data after testing and hyperparameter setting.

- The speed of default vs parameters is negligible for a dataset this size, but scaling up would likely start to be much more of a difference. Random forests with parameters were over 11x more cost intensive than the default, but the results we far better for the tuned version.
- The biggest limitation would be scale and data types. The overfitting on this model might fix itself with a larger dataset but might repeat the same memorization mistakes if the data types remained the same. Another issue would be the increase in costs. Sure it's only fractions of a penny difference to process the parameters on a set this small but if scaled up hundreds of sizes larger it would get far more expensive very quickly.

Future Improvements

- I would try experimenting with other linear models and try to remove some of the more irrelevant features. This may also help with the overfitting issue that the random forest has.
- I think the same dataset but at a much larger scale may have been interesting. A larger dataset could have decided some of the features to drop and even solved some of the overfitting issues.
- I leaned too heavily into the Random Forest not paying attention to the linear relationships. During the EDA process maybe doing some earlier feature importance tests and trying to clean the data to better fit the linear models would have made a difference. It also may have made the results more readable and remove many of the outliers skewing the means of each model.